

# Mobile App Traffic Audit Using Proxy Tool

## Abstract

Mobile applications constantly exchange information with backend servers through network requests. This project demonstrates how a proxy tool can be used to observe and analyze traffic generated by a mobile application in an authorized environment. The project helps students understand request flow, HTTPS interception, API behavior, and common mobile application security considerations.

## Objectives

- To understand the role of proxy tools in network security testing.
- To monitor mobile application traffic using an interception proxy.
- To inspect requests, responses, headers, endpoints, and security-related behavior.
- To identify limitations such as certificate pinning and encrypted payloads.
- To present the findings as an academic mini project.

## Tools Used

- Burp Suite or OWASP ZAP
- Mobile device or emulator
- Wi-Fi network
- Laptop or desktop system
- Proxy CA certificate for HTTPS inspection in a legal and controlled setup

## Methodology

1. Install and launch the proxy tool on the laptop.
2. Connect the mobile device and laptop to the same network.
3. Set the mobile device proxy to the laptop IP address and proxy port.
4. Install and trust the proxy CA certificate when HTTPS inspection is required.
5. Launch the target app and capture the traffic.
6. Analyze methods, headers, payloads, endpoints, and status codes.
7. Record observations and limitations.

## Observations

The captured traffic can reveal HTTP methods, API endpoints, response codes, and data exchange patterns. It may also show authentication headers, session handling mechanisms, and the way the application reacts to network errors. If certificate pinning is enabled, the application may refuse interception and generate connection failures.

## Limitations

- Certificate pinning may block inspection.
- Some apps may use additional application-layer encryption.
- Ethical and legal restrictions apply to unauthorized targets.

## Conclusion

The project provides a practical introduction to proxy-based traffic analysis in mobile app security. It helps students understand how traffic monitoring supports vulnerability identification, backend mapping, and secure development practices.

## Viva Questions

1. What is a proxy tool?
2. Why is a proxy used in mobile traffic auditing?
3. Why must a CA certificate be installed?
4. What is certificate pinning?
5. What type of information can be seen in captured traffic?
6. What are the legal limits of this method?